

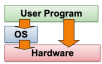
CS2106 Cheatsheet, AY26/27 S1

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1 Introduction

OS is an **intermediary program** between **computer hardware** and **user**.

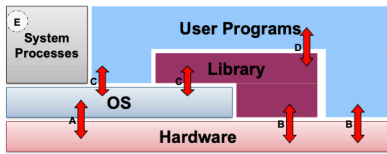
First Computer	OS Type: No OS - Program directly by physically changing the hardware configuration (cables, switches, punched paper tape) Advantage: - Minimal OS overhead Disadvantage: - Not portable (to port, need to manually rewrite the program) and Inefficient - E.g. Electronic Numerical Integrator And Computer (ENIAC), Harvard Mark I	
Mainframes	OS Type: Batch OS - Execute one job at a time. - Support batch processing only - No interactive interface (program via paper tape, magnetic tape, punch card) - User interacts with hardware directly with additional information for OS (e.g. resource required, job specification) Disadvantage: - CPU is idle during I/O from simple batch processing - E.g. IBM 360	
Time-Sharing OS	OS Type: Time-Sharing OS Advantage: - Allow multiple users to interact with the machine using terminals (teletypes) - User job scheduling (illusion of concurrency) - OS manages CPU time, memory and storage - Virtualization of hardware (each program executes as if it has all the resources to itself) - E.g. Apple II PC, IBM PC	
Personal OS	Machine may be dedicated to user, not timeshared between users Windows model: - Single user at a time, but possibly more than 1 user can access Unix model (General time-sharing model): - One user at the workstation, but other users can access remotely	

Motivation of OS

- Abstraction:** Hide low-level details and present higher-level functionality to the user
- Resource Allocator:** Manage all resources (CPU, Memory, I/O) and arbitrate potentially conflicting requests for efficient and fair resource use
- Control Program:** Control execution of programs to prevent errors and improper use of the computer

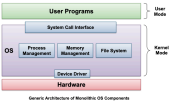
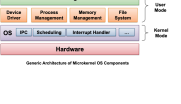
1.1 OS Structures

- OS is software that runs in **Kernel Mode** with direct access to all hardware resources (w/o system calls, normal libraries and 'normal' I/O in kernel mode)
- Other software operates in **User Mode** (i.e. limited/controlled access to hardware resources)



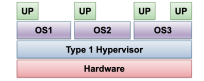
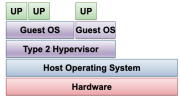
- A:** OS executing machine instructions
- B:** Normal machine instructions executed (program/library code)
- C:** Calling OS using **system call interface**
- D:** User program calls library code
- E:** System processes
 - Provide high-level services, usually part of OS

OS Structures

Monolithic	- Big program (e.g. Linux source code) - If Kernel fails, BSOD - Better performance	
Microkernel	- Smaller and cleaner abstraction - Provides basic and essential facilities: - Inter-Process Communication - Address space management - Thread management - Higher-level OS services are run outside of the kernel, using IPC to communicate. - Lower performance	

1.2 Virtual Machines

A **Virtual Machine** emulates the underlying hardware for running multiple OS, created and managed by a **Hypervisor / Virtual Machine Monitor (VMM)**

Type 1 Hypervisor	Type 2 Hypervisor
	
- Type 1 hypervisor: - AKA bare-metal hypervisor - Runs in host OS - Provides individual virtual machines to guest OSes - E.g., IBM VM/370, VMware ESX	- Type 2 hypervisor: - Runs in host OS - Guest OS runs inside Virtual Machine - E.g., VMware Workstation, VirtualBox, GEMU

2 Process Abstraction

- Process** is the OS abstraction for a running program.

Process Context

Layer	Component	Stores
Memory	Text	program instructions
	Data	global variables and static variables
	Stack	collection of stack frames
	Heap	region of memory used to store dynamically allocated data
Hardware	General-Purpose Registers (GPRs)	temporary data
	Program Counter (PC)	next instruction address

Layer	Component	Stores
	Stack Pointer (SP)	top of stack frame address
	Frame Pointer (FP)	fixed location in stack frame
OS	Process ID (PID)	unique process ID
	Process State	current process state

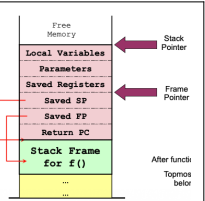
2.1 Memory & Function Calls

- Each function invocation **pushes a new stack frame onto the stack**, while each function return **pops the top stack frame**.
- FP** points to the fixed base of the current stack frame, allowing access of arguments and locals with constant offset.

Stack-Frame Component	Who	Why
Local variables	Callee	Callee's own local variables
Parameters (only those that don't fit in registers)	Caller	so the callee can read the parameters of function call
Saved GPRs	Callee	Copy of GPRs which callee modifies; protect the caller's values across the call
Saved old SP	Callee	restore the caller's SP
Saved old FP	Callee	restore the caller's FP
Return address / Saved PC	Caller	return to the right instruction after the callee returns

Function Call Convention

- Function Call Preparation:**
 - Caller: pass **parameters using registers and/or the stack**
 - Caller: save **return PC on stack**
- Transfer of Control from Caller to Callee:**
 - Callee: save registers used by the callee, old SP and FP
 - Callee: allocate space for **local variables of the callee on the stack**
 - Callee: adjust the SP (and FP) to **point to the new stack top**
- Callee Function Executes**
- Callee Function Returns**
 - Callee: save **return result in return register** (if applicable)
 - Callee: **restore saved registers and SP, FP**
- Transfer of Control from Callee back to Caller using saved PC:**
 - Caller: Use return result and continues execution of the program



Register Saving and Spilling

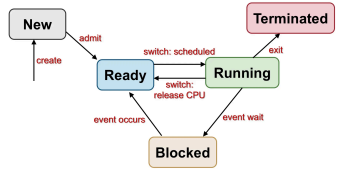
- Register Saving:** caller and callee may use the same registers, so we need to save the registers and restore them after the call.
- Register Spilling:** When a function has more arguments than the number of registers, the extra arguments are spilled to the stack
 - Callee-saved** (default 2106 convention): the callee saves the old values of the registers into its stack frame and restores them after the call
 - Caller-saved:** the caller saves needed registers into its stack frame and restores them after the call

Heap Memory

- heap** is a memory region to store dynamically allocated memory. (e.g. malloc and free in C)
- Dynamically allocated data **cannot be stored in:**
 - Data region:** size must be known at compile time
 - Stack:** the data may outlast the function call

2.2 Process States

Process State Model



Multi-Process Management

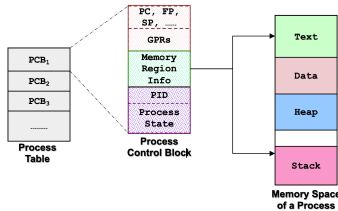
- With 1 CPU core, at most 1 process can be running at a time (2106 convention)
- With m CPU cores, at most m processes can be running at a time

Process Queues

- The OS maintains a queue of processes for each state
 - Ready Queue:** processes that are ready to be scheduled to run on the CPU
 - Blocked Queue:** processes that are waiting for an event to occur
 - may be separate queues for diff types of blocked (e.g. I/O / signal blocked)

2.3 Process Control Block (PCB)

A **Process Control Block (PCB)** or **Process Table Entry** is a data structure that **describes the execution context** for a process, maintained by kernel.



2.4 System Calls

System Call is an OS API for a **user program** to request services in Kernel.

System Call Mechanism

- User invokes the library call (e.g. getpid())
- Library call places the system call number into a register.
- Library call invokes the **trap instruction to switch from user mode to kernel mode**.
- In kernel mode, the **dispatcher identifies the system call and passes control to the appropriate system call handler**.
- System call handler executes the system call
- System call handler ended by restoring CPU state and return to user mode.
- Library call returns the result to the user program.

Exception and Interrupt

- Exception:** Occurs due to program execution, synchronous
 - E.g. divide by zero error or a page fault
- Interrupt:** Occurs independent of program execution, asynchronous
 - E.g. hardware interrupt or a software interrupt

2.5 A Case Study of Processes in Unix

2.5.1 Process Creation: Fork -> Exec

fork()

- int fork()** creates a child process by duplicating currently executing parent.
- Both parent and child continue from **immediately after the fork()** call.

- Child is initially an **almost exact duplicate**:
 - Same code and **initial address-space contents** (conceptually duplicated, **not** shared)
 - Copied register and execution context
 - Different PID and PPID
 - Different `fork()` return value: parent receives the child's PID; child receives 0

Nondeterministic scheduling

- Parent/child order is **nondeterministic**: parent first, child first, or interleaved.

Independent Address Spaces

- Parent and child initially **see equivalent values** after `fork()`, but do **not** share memory.
- Stack, heap, data, and code image are **conceptually duplicated**.
 - Modifying a variable in one process does **not** modify the other.
 - Shared kernel resources are inherited (e.g. opened files), not the address space.

exec() replaces the process image

- `exec()` **replaces** the current process (code and data) with a new executable:
- Begins at the new program's entry point
 - Discards the old stack and execution state
 - **Retains the same PID** and broader process identity
 - **Successful exec() does not return** — the old image no longer exists
 - Variants: `exec(full path, arg0, ..., NULL)`, `execv("/bin/ls", args)`, `execve, execlp(relative path, arg0, ..., NULL)`, `execvp`

main Command-Line Arguments

- ```
int main(int argc, char **argv[], char **vp)
• argc: number of arguments, including the program name
• argv: array of C strings arguments; argv[0] is conventionally the executable name
• vp: array of C strings environment variables
```

**init and Process Tree**

- A process can only be created by forking an existing process, so Unix processes form a **process tree**.
- Root is **init**: created by the kernel during boot, traditionally **PID 1**
  - Common ancestor of user processes; typically spawns OS/system programs
  - Adopts orphaned processes
  - **Cannot be killed** even though it runs in user space; if it crashes → **kernel panic**

2.5.2 Process Termination & Lifecycle

- exit(int status)** terminates the current process. **Does not return**.
- **exit status**: zero if normal/successful termination, non-zero if error/abnormal
  - Returning from `main()` **implicitly** invokes `exit()` with return value becomes the exit status
  - On process `exit()`:
    - Process state is set to **Zombie**
    - Most resources are released (e.g. file descriptors)
    - Some information is **not releasable** (so the parent can `wait()`):
      - PID
      - Exit status
      - Process accounting (e.g. CPU time)

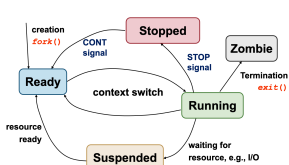
- wait()**
- pid\_t wait(int \*status)** lets a parent synchronise with any child(s):
- **Blocks** until at least one child terminates
  - Returns the PID of a terminated child
  - Stores the child's exit status through `status`, else `NULL`
  - Kernel can write into the parent's memory because it is privileged
  - Variants:
    - `waitpid()` — wait for a specific child
    - `waitid()` — wait for child state changes

**Zombie vs Orphan**

- **Zombie**: child that has **exited** but **has not yet been consumed** by parent via `wait()`
  - Occupies a process-table entry; too many can exhaust the table (older Unix: may need reboot)
- **Orphan**: a **still-running** child whose **parent has terminated**
  - `init` becomes its pseudo-parent
  - When the child later terminates, `init` will clean up with `wait()`
  - If the parent dies while the child is **already a zombie**, `init` reaps that leftover state too

- Parent–Child Lifecycle**
1. Parent fork a child
  2. Child optionally exec a new program
  3. Child `exit` and becomes a zombie
  4. Parent `wait`
  5. Kernel removes the child's remaining process-table entry

**Unix Process States**



3 Process Scheduling

- Concurrent processes** execute during the same period:
- **Virtual parallelism**: pseudo-parallelism.
  - **Physical parallelism**: multiple CPUs/cores execute processes in parallel.

**Time slicing** interleaves processes' instructions and requires a **context switch** at each handover. It works on **single core** (slice instructions) or **multiple processors** (slice instructions and cores).

**Scheduling**

- The **scheduler** uses a **scheduling algorithm** to choose the next process:
- **Non-preemptive (cooperative)**: runs until it blocks or voluntarily yields the CPU.
  - **Preemptive**: runs for a fixed quota, then is suspended.

3.1 Scheduling for Batch Processing

- Batch systems have **no user interaction** and mainly use **non-preemptive scheduling**.
- Criteria:**
- **turnaround time** = finish - arrival
  - **throughput** = tasks finished per unit time
  - **waiting time** = total time waiting since creation
  - **CPU utilization** = percentage of time the CPU runs a task.

**First Come First Serve (FCFS)**

- Uses a **FIFO queue**. Its head runs until **completion or blocking**; a blocked task leaves the queue and rejoins its **tail** when ready.
- **Traits: non-preemptive, no starvation.**
  - **Costs: the convoy effect** prevents CPU- and I/O-bound processes from running concurrently; **reordering** tasks can lower average waiting time.

**Shortest Job First (SJF)**

- Non-preemptively selects the task with the **smallest total CPU time**.
- Requires **advance knowledge** of each task's total CPU time.
  - **Minimizes average waiting time**, but its bias toward short jobs can **starve long jobs**.

**Predicting CPU Time**

- Estimate the next CPU-bound phase from previous phases:
- $$\text{Predicted}_{n+1} = \alpha \text{Actual}_n + (1 - \alpha) \text{Predicted}_n$$
- `Actualn`: most recent CPU time; `Predictedn`: past estimate;  $\alpha$ : weight on the recent event; `Predictedn+1`: new estimate.

**Shortest Remaining Time (SRT)**

- Preemptively runs the job with the **shortest remaining (or expected) time**; a newly arrived shorter job **preempts** the running job.

3.2 Scheduling for Interactive Environment

- Interactive systems mainly use a **periodically invoked, preemptive scheduler** for good **response time** and **predictability**.
- Response Time**: First CPU time - time task creation

- Timer & Time Quantum**
- **Timer interrupt interval (ITI)**: interval between interrupts that invoke the OS scheduler; typically **1–10 ms**.
  - **Time quantum**: process execution allowance; a **multiple of the ITI**, constant or process-dependent, typically **5–100 ms**.

**Round Robin (RR)**

- Preemptive FCFS** using a **FIFO queue**. Its head runs until its **quantum expires, it finishes, or it blocks**; an unfinished task rejoins the tail, as does a blocked task once ready.
- **Timer interrupts** detect quantum expiry.
  - **Wait bound**:  $(n - 1)q$  before getting the CPU, for  $n$  tasks and quantum  $q$ .
  - **Large  $q$** : higher CPU utilization, longer waits. **Small  $q$** : more overhead, shorter waits.

**Priority Based**

- Runs the task with the **highest priority value**.
- **Preemptive**: a higher-priority arrival preempts a lower-priority running process.
  - **Non-preemptive**: a late higher-priority arrival waits for the next scheduling round.
  - **Starvation**: low-priority processes may never run. Mitigate by **lowering the running process's priority** after each quantum, or **excluding it from the next round** after giving it a quantum.

**Multi-Level Feedback Queue (MLFQ)**

**Higher priority wins; equal priorities use RR**. New job → highest priority, Deplete time quantum → priority decrease, Job gives up → priority maintained. This minimizes **response time for I/O-bound processes** and **turnaround time for CPU-bound processes**.

**Lottery Scheduling**

Distribute **lottery tickets** for resources (CPU time, I/O devices). At each decision, **randomly choose** an eligible ticket; in the long run, **X% of tickets gives X% resource use**.

4 Inter-Process Communication

- Inter-Process Communication** transfers information between processes.
- Shared-Memory**
- A **shared memory** region  $M$  is created (`shmget`) and attached (`shmat`) by the processes  $P_1$  and  $P_2$ , enabling communication.
- Efficient: OS only needs to Create and Attach memory region
  - `shm[0]` to declare if the value is ready
  - `shmget` to create, `shmat` to attach, `shmdt` to detach and `shmctl` to destroy

**Message Passing**

- Process  $P_1$  prepare messages  $M$  (stored in **kernel memory space**) and send to  $P_2$  using **system calls**. Requires extra properties like Name and Synchronization.
- **Direct Communication**: Sender and Receiver of message explicitly name the other party (ie. unix domain socket). One-to-one communication between processes.
  - **Indirect Communication**: Messages are sent / received from message storage known as mailbox or port (i.e. unix message queue). Many-to-Many communication between processes.
  - **Blocking Primitives (synchronous)**: Receiver is blocked until message has arrived.
  - **Non-Blocking Primitives (asynchronous)**: Receiver either receive the message if available or some indication that message is not ready.

4.1 Unix Pipes and Signals

- A process has 3 default communication channels: `stdin (scanf)`, `stdout (printf)`, `stderr`. Declared in bash with "`"`".
- ```
int pipe(int fd[]) functions as circular bounded byte buffer (writers wait when buffer is full) and implicit synchronization (readers wait when buffer is empty)
```
- Returns `-1` if creation of `fd` fails
- Depending on Unix version, pipes may be:
- Half-duplex: unidirectional with one write end and one read end
 - Full-duplex: bidirectional with any end for read and write
- `pipe(fd[2])` creates an array of file descriptor with `READ_END=0`, `WRITE_END=1`.
 - process may `write(fd[WRITE_END], str, strlen(str)+1)` or `read(df[READ_END], buf, sizeof(buf))` from pipe

- `dup2(fd[0], STDOUT_FILENO)` and `dup2(fd[1], STDOUT_FILENO)`
- Unix Signal**
- signal sent to a process using an asynchronous notification regarding an event.
- Eg. Kill, Interrupt, Stop, Continue, Memory Error, Arithmetic Error...

4.2 Threads

Multiple **threads** run concurrently in a process, sharing the **same memory context** (e.g. text, data, heap) and **OS context** (PID and other resources), **differing** only in **hardware context** (registers and stack).

User and Kernel Thread

- **User Thread** is thread implemented as user library and handles by its runtime process, not aware by kernel.
 - Pros: More configurable and portable as thread operation are just library calls.
 - Cons: Scheduling performed at process level (1 thread blocks out whole process); cannot exploit multiple CPU.
- **Kernel Thread** is thread implemented by OS, handled as system calls, enabling thread-level scheduling.
 - Pros: More than 1 thread in same process can run on multiple CPUs as it is handled by Kernel
 - Cons: Thread operation with system call (higher overhead), and generally less flexible.
- **Hybrid Thread** uses both user thread and kernel thread at the same time.

5 Synchronization

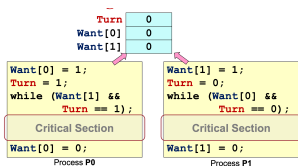
- Concurrent execution is **non-deterministic**, so a **race condition** occurs when the outcome depends on the order of shared-resource access/modification.
- Critical section (CS)** is code segment that only one process may execute at a time with the following properties
- **Mutual Exclusion**: Only one processes can executes CS.
 - **Progress**: If no process in CS, one of the waiting processes is granted access.
 - **Bounded Wait**: Upperbound time for P_i in CS, before other process can enter.
 - **Independence**: Process in non-CS should never block other process.

- Outcome of incorrect CS implementation:
- **Deadlock**: All processes blocked and stops execution
 - **Livelock**: Result of deadlock avoidance mechanism, where processes keep changing state and make no progress
 - **Starvation**: Some processes are blocked forever

Low-Level Implementation: TestAndSet

- `TestAndSet` Register, `MemoryLocation` loads the content from memory to register, and stores 1 to memory, happens on hardware level (atomic).
- ```
EnterCS(int* Lock) while (TestAndSet(Lock) == 1) // Loop -> unlock
ExitCS(int* Lock) *Lock=0; // Set lock to 0
```

**Higher level Language: Peterson's Algorithm**



**Higher level synchronization: Semaphore**

- Semaphore is an integer  $S \geq 0$ , represent number of jobs that can run concurrently (counting semaphore). mutex when  $S \in \{0, 1\}$ .
- `Wait(S)`: When  $S=0$ , blocks and puts process to sleeping queue. Else,  $S- (P())$ , `Down()`
  - `Signal(S)`: Wakes up 1 sleeping processes if any. Else,  $S++ (V())$ , `Up()`
- $$N_{CS} = \#Signal(S) - \#Wait(S) \quad S_{current} = 1 + N_{CS}, \quad N_{CS} \leq S_{initial}$$